DATA ARCHIVING AND PRESERVATION OF DATA ON HEALTH – A NEW COLLABORATION BETWEEN THE DATA ARCHIVES AND SERVICES FROM SWEDEN, NORWAY, FINLAND, AND DENMARK

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Bergen NORWAY:
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Gothenburg SWEDEN:
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Odense DENMARK:
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Tampere FINLAND:
FSD

CURRENT STATUS FOR UNITS FOR HEALTH DATA

- Denmark:
  Unit for health data since 1996
- Sweden:
  Unit for health data since 2010
- Norway:
  No special unit for health data
- Finland:
  No special unit for health data

THE NEW NORDIC COLLABORATION

Earlier collaboration between DDA and SND

The meeting in Odense in November 2012 also included FSD and NSD:

Bodil Stenvig (DDA),
Martin Brandhagen (SND),
Henrik Sejersen (DDA),
Vigdis Kvalheim (NSD),
Mari Kleemola (FSD),
Gry Henriksen (NSD)
Heidi Wittendorff Jensen (DDA),
Arja Kuula (FSD),
Elisabeth Strandhagen (SND)

(… and H. C. Andersen from Odense…)

DDA - DANISH DATA ARCHIVE

- DDA was established in 1973 as a data service for researchers and students

- It is a unit of the National Danish Archives and is now a Research Infrastructure (RI) for survey data from three fields:
  - Social science
  - Health science
  - History

DDA - DANISH DATA ARCHIVE

- The archival act in Denmark gives permission to the National Archives to archive and maintain data and documents containing personal identifiers

- And an agreement with the Danish Data Protection Agency DDA preserves data materials containing personal identifiers.

- Data and the personal identifiers are stored separately and a special permit is required for access to the data.

DDA HEALTH

- More than 500 studies have been archived in DDA Health
- A proximately 50% of these include personal identifiers
- DDA Health receives approximately 60 studies per year
- Stimulation of voluntary archiving by newsletter, presentations, undergraduate courses and research training courses, network of health and medical researchers, supporting data processing and documentation
- Recycling of data and metadata increases remarkably

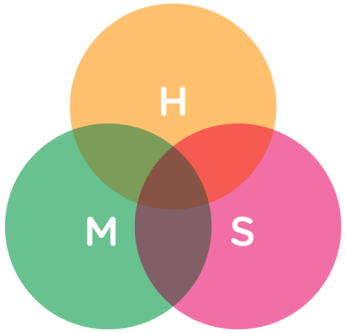
The aim is to be one of the windows to the global research data arena – and the Danish Ministry of Sciences, Technology and Innovation expects DDA Health to contribute by qualifying the use of research data in Denmark

SND – SWEDISH NATIONAL DATA SERVICE

SND was initiated by the Swedish Research Council, established 2008

It is hosted by the University of Gothenburg which earlier hosted the Swedish social science data service (SSD) since 1980

SND is a service organisation for Swedish research within
- social sciences
- medicine and health sciences
- humanities



SND gives access to existing data within and outside Sweden and gives support and guidance to researchers throughout the whole research process.

SND AND HEALTH

Current situation: SND does not handle personal data

April 2012: The Data Inspection Board (DI) stated that SND's activities violates the Personal Data Act and has to stop handling personal data.

June 2012: The University of Gothenburg appealed the decision from the Data Inspection Board, where SND was submitted to cease collecting and managing research material containing personal data. The University stated that the decision is based on an overly general assessment that does not specify exactly what personal data processing operations conflicts with the Personal Data Act. GU also claimed that the decision from the Data Inspection Board does not distinguish between SND's overall mission and the actual personal data processing operations.

September 2012: The Data Inspection Board maintained it's decision regarding SND

May 2013: We are still waiting for a decision…

SND has data or description for 57 studies within the health area.

NSD – NORWEGIAN SOCIAL SCIENCE DATA SERVICE

NSD was established 1971 and is since 2003 organized as a company owned by the Ministry of Education.

NSDs mandate – Interdisciplinary, main focus on social sciences, also humanities, and health sciences:

- **Research Council of Norway**
  NSD is the RCN's main instrument building various national support or infrastructure services, such as a national archive for research data and as national system for remote access to statistical and administrative data
- **Data Protection Agency**
  NSD as a broker between the Agency and the scientific community. Today NSD function as Data Protection Official for research
- **Statistics Norway**
  Since 1976, a formal agreement on dissemination of data for research purposes. Allowing for extensive use of aggregated and individual data; for instance surveys and register data
- **National Archive** from informal to formal cooperation in 2013

NSD AND HEALTH

Health Data from Statistics Norway's Surveys:
The collaboration with SN is unique in the international context. The use of NSD for managing data has resulted in the establishment of procedures for safeguarding the protection of personal data while allowing for relatively extensive use of individual data for research purposes.

Health Data from Registers

Health Data from Norwegian Research Projects
NSD archives data from health research projects that are subject to legal deposit as a result of RCN's contractual terms and conditions.
P.t. 2,500 questions from 250 projects are about health (incl. surveys from Statistics Norway)

FSD – FINNISH SOCIAL SCIENCE DATA ARCHIVE

- FSD is a national resource centre for social science research and teaching
- FSD archives, promotes and disseminates digital research data for research, teaching and learning purposes. The archive is funded by the Ministry of Education and Culture and is a separate unit of the University of Tampere
- FSD promotes open access to research data, and the transparency, accumulation and efficient reuse of scientific research.
- Majority of all datasets (1300+) from social sciences including qualitative data (10%)
- FSD tries to get a new and larger national mandate – broaden services to health sciences, humanities and behavioral sciences

FSD AND HEALTH

- Topical classification of datasets ”health sciences”: 125 datasets (<10% of all)
- PI/data depositor from health science department/organisation: < 40 datasets
- Users: < 30 researchers from health science departments/organisations (usually order several datasets at once)

A new application for funding period 2014-2015 includes “Charting the Prerequisites of Extending FSD Services to Health Research”

- exploring the strategies for archiving data in health sciences
- affecting ethical practices of data management in health research
- concrete pilot of archiving health data
- studying the implementation of DDI metadata schema for health data

THE NEW NORDIC COLLABORATION – THE OBJECTIVES OF THE MEETING IN ODENSE

- Focus on the current legal situation on ‘data protection’ in the four countries
- Current situation on archiving, preserving and dissemination of data from the medical and health sciences
- The Nordic register research tradition within epidemiology can be stronger with more data from the medical research communities
- To learn from each other in the Nordic data services

THE NEW NORDIC COLLABORATION – LONG TERM OBJECTIVES

- Adding value to the already existing research infrastructures in Europe and globally
- Improving collaboration and knowledge on data sharing in the Nordic counties and abroad
- Supporting the development of units for medical science data in the data archives and data services in all the Nordic Countries

VISION:

The Nordic countries should be a region with common 'love' for data processing of personally identifiable / person identifiable data such as population studies and population registers in the Nordic countries

It will provide great benefits for the social sciences and medical research e. g. opportunities to understand and analyze cohorts, time series and the consequences of 'society' and social change.

NORDFORSK

… an organisation under the Nordic Council of Ministers

Is working to make Scandinavia to a 'region' with common framework for data preservation and data access and analysis of data from the Nordic countries – that means no limit to data transfer across the Nordic countries, etc. ..

Currently there are routines for delivering survey data from e.g. DDA to a Swedish researcher - after the otherwise applicable extradition procedure.

CHALLENGE: GOOD DATA MANAGEMENT

- To incorporate good 'data management' including giving free access to the data collected
- Getting researchers to deposit data to data archives
- Getting the common tools that can be used for data documentation
- Working together across research infrastructures takes time

CHALLENGE 1: MEDICAL TERMINOLOGY

Development of DDI-L and CESSDA subject categories for documentation of epidemiological and clinical databases and clinical trials / research, ie. development of data archives data management

There is a need for a more specific and broadened terminology in the standards for metadata documentation

CHALLENGE 2: DOCUMENTATION OF CLINICAL DATA

Establish good research documentation to ensure that the research material and results in a fast and easy way can be reproduced, controlled and re-used

Key factors for successful research documentation:
- a well thought-out folder structure
- analysis plan
- log book
- codebook
- …

CHALLENGE 3: DATA SHARING WITHIN THE MEDICAL FIELD

- Legislation
- Lack of a data sharing culture (maybe partly because of…)
- Lack of data citation practices
- Much focus on number of publications
- Stronger “competition” between research groups
- Lack of metadata and documentation

CHALLENGE 4: USE OF DATA FROM HEALTH REGISTERS AND HEALTH QUALITY REGISTERS

- Institutional and legal bottlenecks
- Technical bottlenecks
- Financial impediments
- Linguistics impediments
- Data quality and coverage problems
- Qualifications and experience

CHALLENGE 4 CTD: USE OF DATA FROM HEALTH REGISTERS AND HEALTH QUALITY REGISTERS – ACROSS THE BORDERS

- Structural problems (institutions, people, policies and practices and capital resources) restrict opportunities for more research and thus the progress of science and society
- Different legislation in the Nordic countries regarding personal data – and between different authorities
- A harmonised European legal framework needed – political will to use strong incentives, changing legislation and establishing new funding schemes to support and enforce data access and exchange

BENEFITS

- That we can put resources together
- That we can communicate data across the countries and make them visible

NEXT STEP

- Information stand in the EuroEpi 2013 conference in Århus, Denmark (August) www.euroepi2013.org
- Medical terminology – ongoing work
- Waiting for the EU data protection legislation process…